

HINT-Blimp: Human INTent Inference from Multimodal Cues for Robotic Blimps

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Video :



Motivation

Can a simple push or spoken command convey high-level intent in robot navigation tasks without menus, tablets, or teleoperation?

Challenges:

- Short interactions are ambiguous
- Many goals may lie in one push direction
- Cues give a direction, not a full trajectory

Key Idea and mathematical formulation

We model human intent as a dynamical system and infer it from sparse cues.

$$\dot{\mathbf{x}} = \mathbf{A}(\mathbf{x} - \mathbf{x}^*)$$

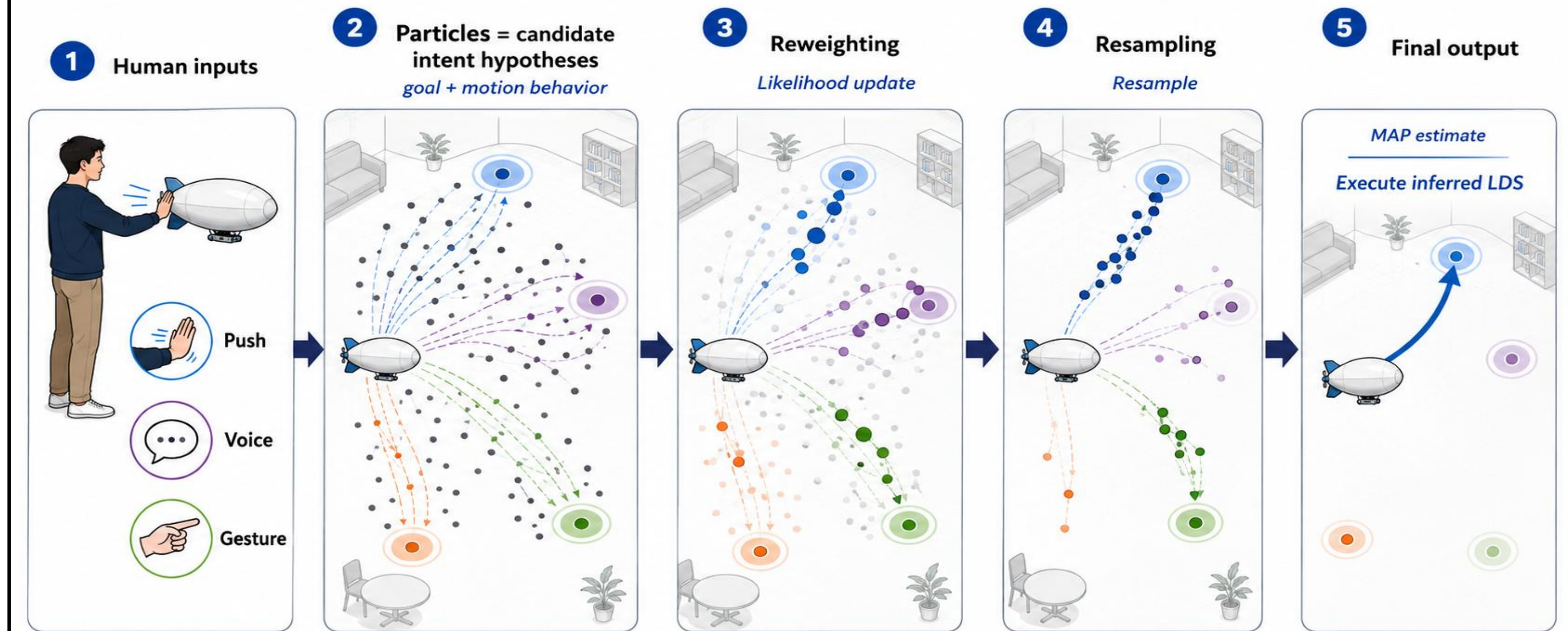
- $\mathbf{x}^* \rightarrow$ where to go(goal)
- $\mathbf{A} \rightarrow$ how to get there(motion style)

Each particle is one candidate $(\mathbf{x}^*, \mathbf{A})$. Cues reweight the belief:

Push $\rightarrow$ velocity alignment with the particle's DS

Voice / gesture $\rightarrow$ bearing alignment with the commanded direction

Particle filter for intent inference



Experiment setup

SBlimp: Crazyflie 2.1 with tilted rotors on a Mylar helium balloon
 OptiTrack motion capture; commands via crazyswarm
 Only the velocity controller runs onboard

Sensing the human:

Pushes: detected from mocap when speed exceeds the threshold velocity

Voice: laptop mic $\rightarrow$ Vosk (straight / right / left / back)

Gesture: webcam $\rightarrow$ MediaPipe Hands (pointing ray)

Particle filter in NumPy on CPU

Experiment Protocol :

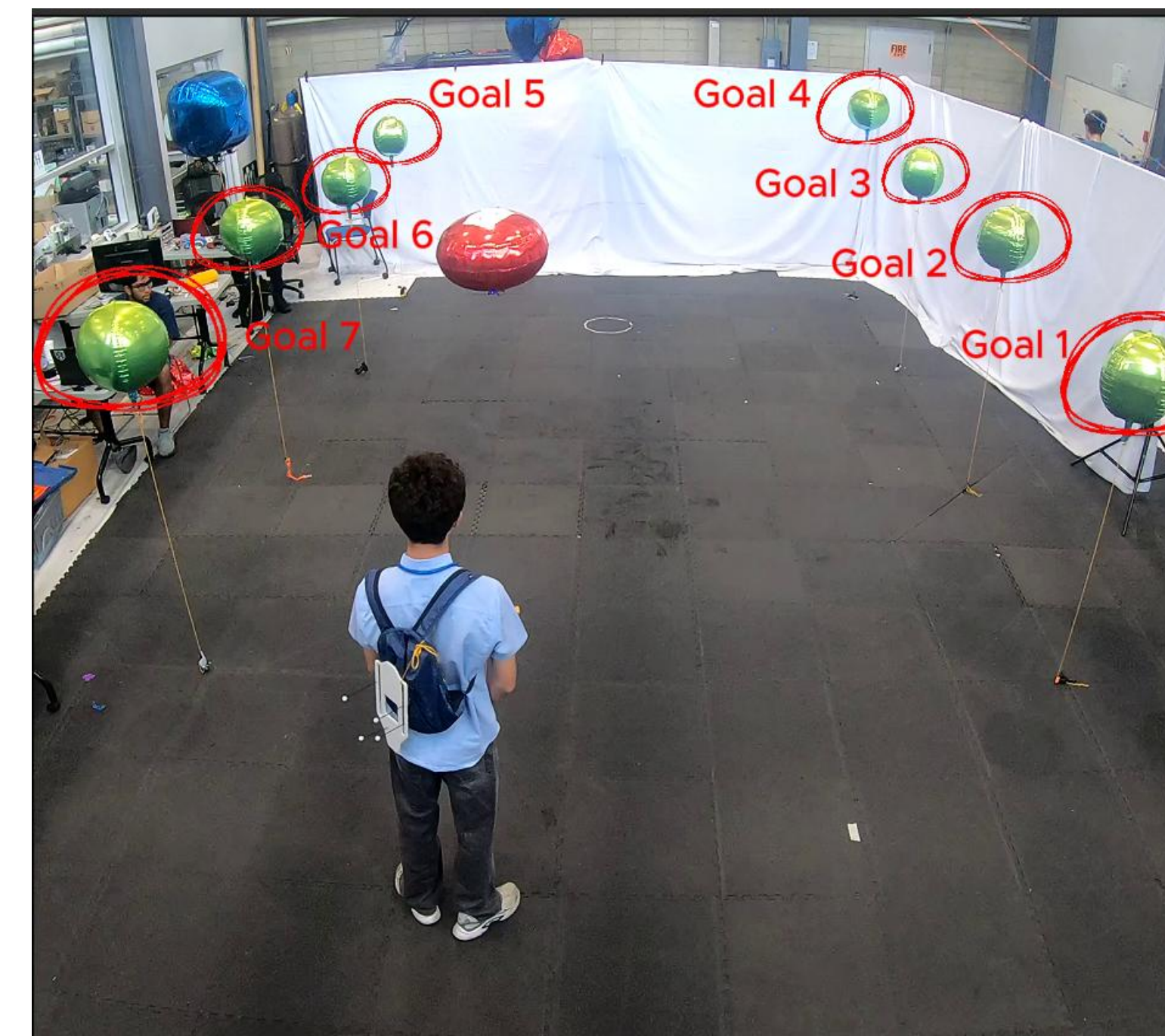
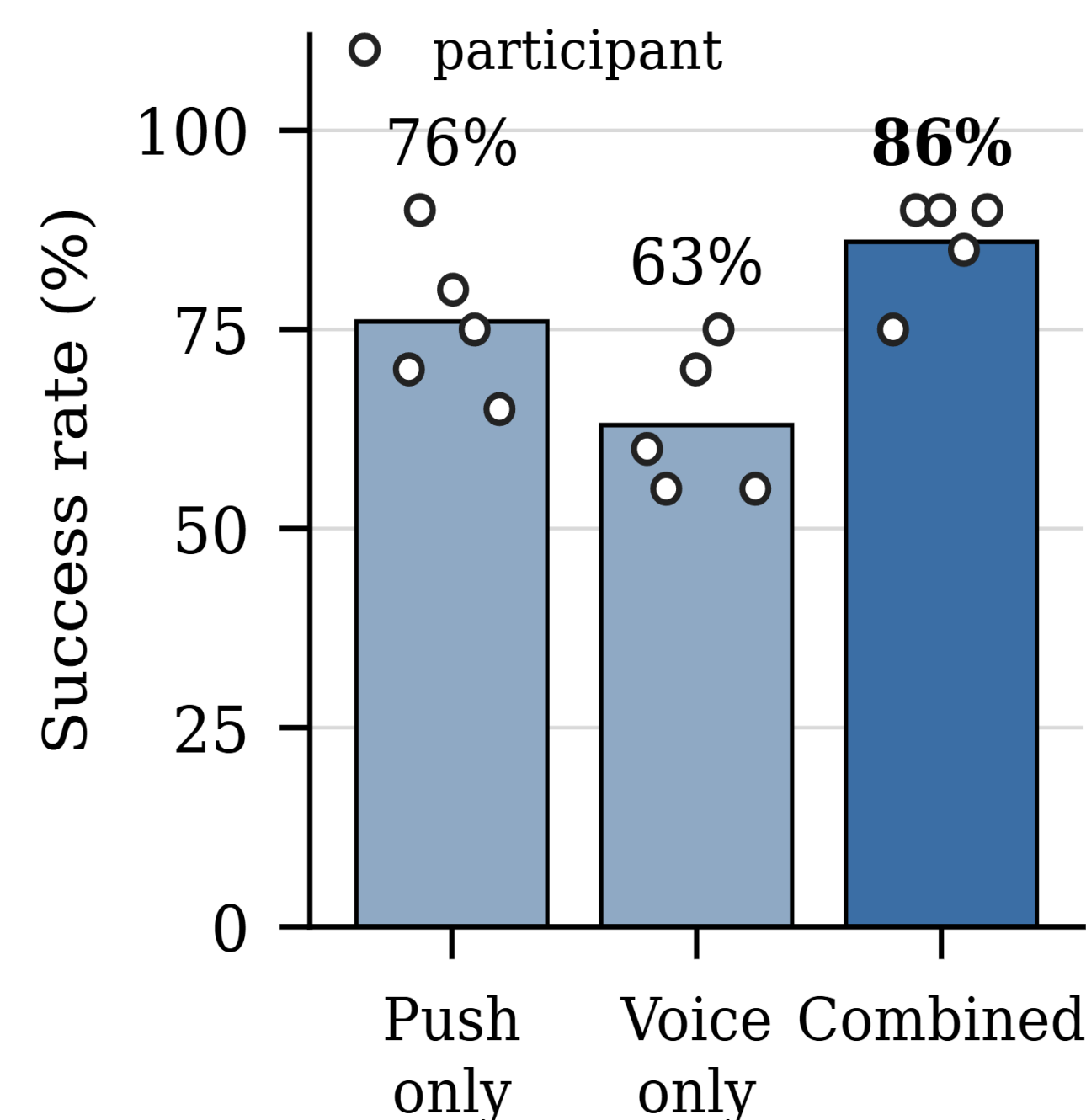
5 participants $\times$ 60 trials = 300 trials

8 candidate goals, one chosen at random per trial, unknown to the robot

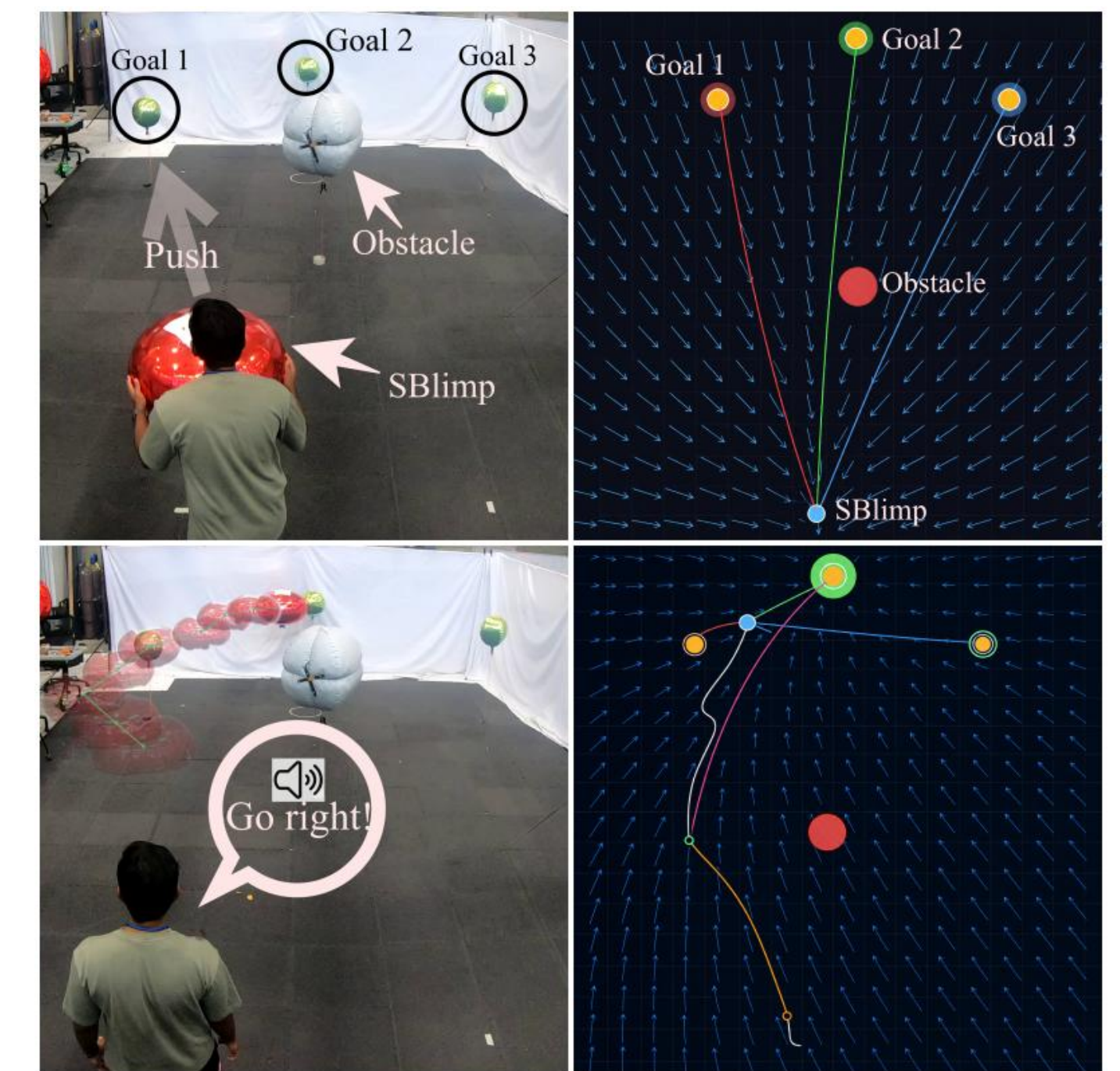
3 conditions: push only / voice only / combined

Success = reaches the correct goal in under 6 interactions

Uniform scene prior, so belief comes entirely from human cues



Experiment 1: The subject has to send the robot to one randomly chosen goal among 8 candidate goals. Combining modalities wins. 86% goal identification vs. 76% push-only and 63% voice-only. Circles are individual participants, combined is also the most consistent across users.



Experiment 2: Steering around an obstacle the robot can't see: The human wants Goal 2, but a straight line hits the obstacle. A push toward Goal 1 (top) leaves all goals plausible; "Go right!" (bottom) reweights toward Goal 2, and the executed trajectory curves around.